INCH-POUND
MIL-DTL-24643/72A
28 January 2010
SUPERSEDING
MIL-DTL-24643/72
9 October 2008

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO + 90 °C, 1000 VOLTS, TYPE LSFCF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

Components 1-4:

First	-	Copper conductor, tin-coated, 4 AWG 420 strands of 30 AWG.
Second	-	Flexible silicone insulation. Diameter: 0.380±0.010 inch
Third	-	Insulation covering - flexible thermoset insulation. Color black. Diameter: 0.450±0.010 inch. Printed white identification marked at a maximum distance of 3 inches, component 1-black, component 2-white, component 3-red, component 4-green.

Components 5-8:

First	-	Copper conductor, tin-coated, 16 AWG 19 strands of 29 AWG.
Second	-	Thermoset insulation. Color black. Diameter: 0.090±0.003 inch. Printed white identification marked at a maximum distance of 3 inches, component 5-orange, component 6-blue, component 7-white-black, component 8-red-black.

Assembly:

Fourth	-	Components 1-4 and 5-8 shall be cabled together with a maximum lay of 11 inches and as follows: Component 5 shall be placed in the outer interstices between component 1 and 2. Component 6 shall be placed in the outer interstices between component 2 and 3. Component 7 shall be placed in the outer interstices between component 3 and 4. Component 8 shall be placed in the outer interstices between component 4 and 1. Fillers may be used as necessary to form a firm well-rounded assembly.
Fifth	-	Optional binder.
Sixth	-	Braided shield of 34 AWG tin-coated copper. Braid angle of 40 to 50 degrees; minimum coverage of 90 percent.
Seventh	-	Optional binder.
Eighth	-	Flexible cross-linked polyolefin jacket with a nominal wall thickness of 0.075 inch (see table I). Color black.

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TABLE I. Construction details.

Military part no. M24643/72	Type & size	Minimum bend radius (inch)	Overall diameter (inches)	
			Type LSFSU	
			(min.)	(max.)
-01UO	LSFCF-42	7.5	1.250	1.350

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (4 AWG) (ohms/1000 feet at 25 °C, max.)	0.298
Conductor resistance (16 AWG) (ohms/1000 feet at 25 °C, max.)	4.62
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	3000
Conductor to shield	3000
Insulation resistance (megohms/1000 feet, min.)	See table I
Conductor to conductor	500
Conductor to shield	250
Jacket flaws	No failure
Conductor and shield continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Thermoset proof test (percent, max.)	
Insulation	50
Jacket	50
Cold bending, cable mandrel diameter 12 times cable diameter	No damage
Bending endurance (25±2 °C, cycles, min.)	25
Physicals (unaged)	
Insulation (flexible silicone 4 AWG and thermoset insulation 16 AWG)	
Tensile strength (lb/in^2, min.)	700

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Elongation (percent, min.)	250
Insulation covering (4 AWG only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	200
Durometer (Type A hardness, min./max.)	60/75
Jacket (overall cable)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	200
Durometer (Type A hardness, min./max.)	60/75
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (inner layer)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Insulation covering (4 AWG only)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Jacket (overall cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Heat distortion (percent of unaged, max.)	30
Permanence of printing (insulation) (cycles, min.)	75
Shrinkage	No failure
Shield - conformance to material (construction and coverage)	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2

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Fillers	2
Insulation	2
Halogen content (percent, max.)	
Insulation	0.2
Jacket	0.2
Fillers	0.2
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	35
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Durometer (jacket) – Type A (hardness, min.)	80
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSFCF	250

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-090)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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